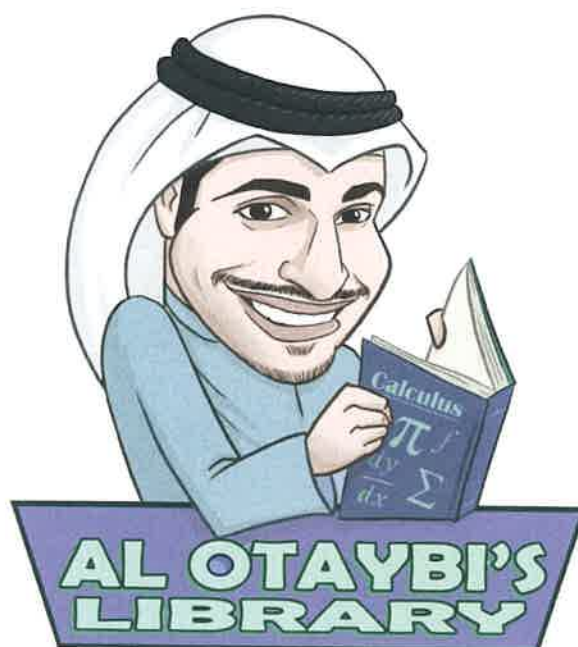


Kuwait University

2021/2022

Fall course



Calculus A

Final

Sections : 5.2 (Part c) - 5.5

5.2 The Definite Integral.

5.5 The Substitution Rule.

(19)

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Riemann Sum



$$R_n = \sum_{i=1}^n f(\bar{x}_i) \Delta x \quad x \in [a, b]$$



خطوات الحل





$$\Delta x = \frac{\text{بداية - نهاية}}{\text{عدد الفترات}}$$

(1) إيجاد Δx

$$\Delta x = \frac{b - a}{n}$$

(2) وضع الصندوق والالتزام بالـ Δx

(3) تطبيق القانون بحسب الحالة

$$R_n = [f(x_1) + f(x_2) + f(x_3) + f(x_4) + \dots] \Delta x$$

Right end point

Left end point

$$\bar{x}_i^* = a + i \Delta x$$

$$\bar{x}_i^* = a + (i - 1) \Delta x$$



Mid point



If $f(x) = x^2$ for $0 \leq x \leq 3$

Evaluate the Riemann sum $R_3 = \sum_{i=1}^3 f(x_i^*) \Delta x$ with

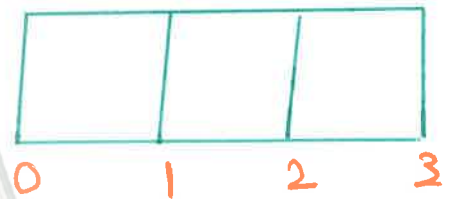
$n = 3$ by taking the sample points x_i^* to be

a) The right end-point

b) The left end point

$f(x) = x^2$ $[0, 3]$ $n = 3$

$$\Delta x = \frac{b-a}{n} = \frac{3-0}{3} = \frac{3}{3} = 1$$



$$\begin{aligned} \text{(a)} \quad R_3 &= (f(3) + f(2) + f(1)) \Delta x \\ &= (9 + 4 + 1) \cdot (1) \\ &= 14 \cdot (1) = 14 \end{aligned}$$

$$\begin{aligned} f(0) &= 0^2 = 0 \\ f(1) &= 1^2 = 1 \\ f(2) &= 2^2 = 4 \\ f(3) &= 3^2 = 9 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad R_3 &= (f(0) + f(1) + f(2)) \Delta x \\ &= (0 + 1 + 4) (1) \\ &= 5 (1) \end{aligned}$$

If $f(x) = 4 - x^2$ for $-2 \leq x \leq 2$

Evaluate the Riemann sum $R_3 = \sum_{i=1}^3 f(x_i^*) \Delta x$ with

$n = 4$ by taking the sample points x_i^* to be

a) The right end-point

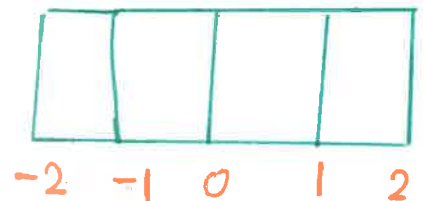
b) The left end point

$$f(x) = 4 - x^2 \quad [-2, 2] \quad n = 4$$

$$\Delta x = \frac{b-a}{n} = \frac{2 - (-2)}{4} = \frac{4}{4} = 1$$

$$\begin{aligned} \text{a) } R_4 &= (f(2) + f(1) + f(0) + f(-1)) \Delta x \\ &= (0 + 3 + 4 + 3) (1) \\ &= 10 (1) = 10 \end{aligned}$$

$$\begin{aligned} \text{b) } R_4 &= (f(-2) + f(-1) + f(0) + f(1)) \Delta x \\ &= (0 + 3 + 4 + 3) (1) \\ &= 10 (1) = 10 \end{aligned}$$



$$f(-2) = 4 - 4 = 0$$

$$f(-1) = 4 - 1 = 3$$

$$f(0) = 4 - 0 = 4$$

$$f(1) = 4 - 1 = 3$$

$$f(2) = 4 - 4 = 0$$

By using Riemann sum approximate $\int_0^{16} (16 - x) dx$

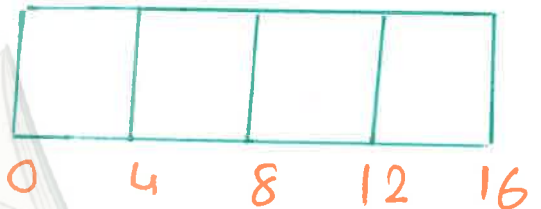
using 4 subintervals taking the sample point to be

- a) The right end-point
- b) The left end point
- c) The mid point

$$f(x) = 16 - x \quad [0, 16] \quad n = 4$$

$$\Delta x = \frac{b - a}{n} = \frac{16 - 0}{4} = 4$$

$$\begin{aligned} \text{a) } R_4 &= (f(16) + f(12) + f(8) + f(4)) \Delta x \\ &= (0 + 4 + 8 + 12)(4) \\ &= 24(4) = 96 \end{aligned}$$

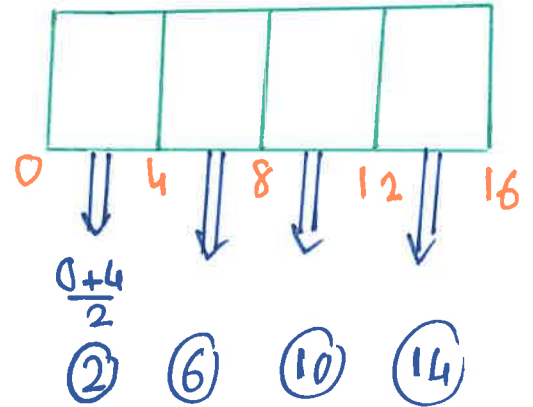


$$\begin{aligned} \text{b) } R_4 &= (f(0) + f(4) + f(8) + f(12)) \Delta x \\ &= (16 + 12 + 8 + 4)(4) \\ &= 40(4) = 160 \end{aligned}$$

$$\begin{aligned} f(0) &= 16 - 0 = 16 \\ f(4) &= 16 - 4 = 12 \\ f(8) &= 16 - 8 = 8 \\ f(12) &= 16 - 12 = 4 \\ f(16) &= 16 - 16 = 0 \end{aligned}$$

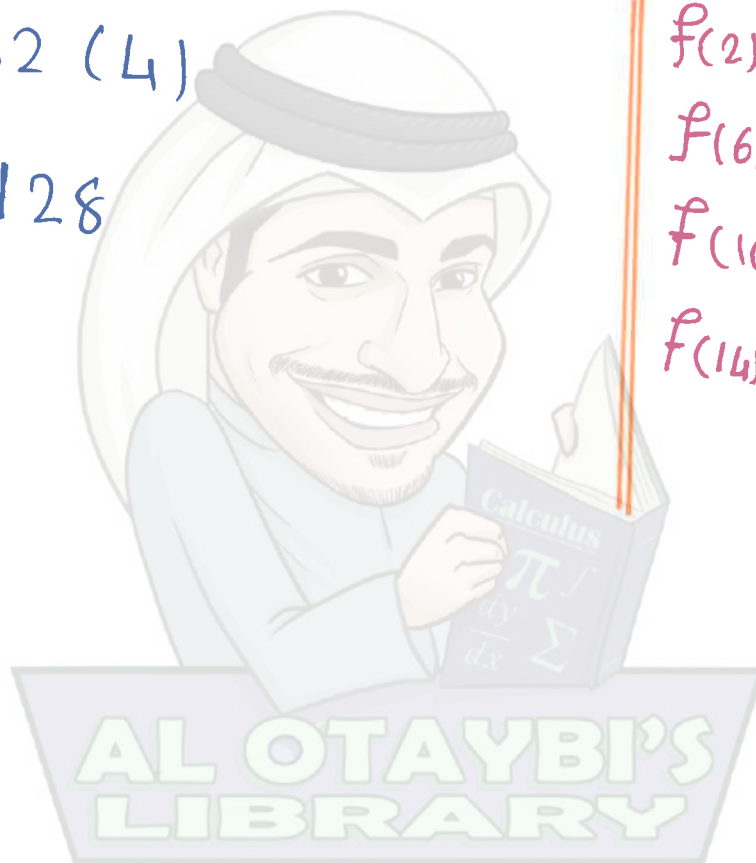
$\Rightarrow$

(c) From mid point



$$\begin{aligned} R_4 &= (f(2) + f(6) + f(10) + f(14)) \Delta x \\ &= (14 + 10 + 6 + 2) (4) \\ &= 32 (4) \\ &= 128 \end{aligned}$$

$$\begin{aligned} f(2) &= 16 - 2 = 14 \\ f(6) &= 16 - 6 = 10 \\ f(10) &= 16 - 10 = 6 \\ f(14) &= 16 - 14 = 2 \end{aligned}$$

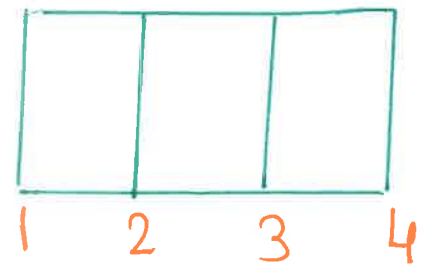


Evaluate the Riemann sum for $f(x) = 3 - x^2$
taking the sample points to be right endpoints and
 $a = 1, b = 4$, and $n = 3$.

Dec 2017

$$f(x) = 3 - x^2 \quad [1, 4] \quad n = 3$$

$$\Delta x = \frac{4 - 1}{3} = \frac{3}{3} = 1$$



$$R_3 = (f(4) + f(3) + f(2)) \Delta x$$

$$= (-13 - 6 - 1)(1)$$

$$= -20$$

$$f(4) = 3 - 16 = -13$$

$$f(3) = 3 - 9 = -6$$

$$f(2) = 3 - 4 = -1$$

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The Riemann sum of $f(x) = x^2$ on the interval $[1, 5]$ using $n = 4$ and taking the sample points to be the right endpoints is :

a) 25 b) 54 c) -30 d) 0 e) None of the above.

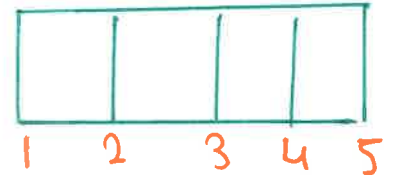
Aug 2017

$$f(x) = x^2 \quad [1, 5] \quad n = 4$$

$$\Delta x = \frac{5-1}{4} = \frac{4}{4} = 1$$

$$\begin{aligned} R_4 &= (f(5) + f(4) + f(3) + f(2)) \Delta x \\ &= (25 + 16 + 9 + 4) (1) \\ &= 54 \end{aligned}$$

(b) ✓



$$\begin{aligned} f(5) &= 25 \\ f(4) &= 16 \\ f(3) &= 9 \\ f(2) &= 4 \end{aligned}$$

Find Riemann sum of $f(x) = \sin x$, $0 \leq x \leq \pi$ with $n = 3$ taking the sample points to be the right endpoints is equal to :

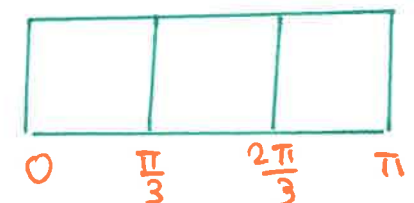
a) 0 b) $\frac{\pi}{3}$ c) $\frac{\pi}{\sqrt{3}}$ d) $\frac{\pi(1+\sqrt{3})}{2}$ e) None of the above.

July 2018

$$f(x) = \sin x \quad [0, \pi] \quad n = 3$$

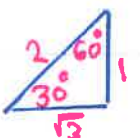
$$\Delta x = \frac{\pi - 0}{3} = \frac{\pi}{3}$$

$$\begin{aligned} R_3 &= (f(\pi) + f(\frac{2\pi}{3}) + f(\frac{\pi}{3})) \Delta x \\ &= (0 + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}) \frac{\pi}{3} \\ &= \frac{2\sqrt{3}}{2} \frac{\pi}{3} \\ &= \sqrt{3} \frac{\pi}{3} = \frac{\pi}{\sqrt{3}} \end{aligned}$$



$$\begin{aligned} f(\pi) &= 0 \\ f(\frac{2\pi}{3}) &= \sin \frac{2\pi}{3} = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \\ f(\frac{\pi}{3}) &= \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \end{aligned}$$

(c) ✓



5.5



Substitution



طريقة تستخدم لتحويل التكاملات المعقدة الى تكاملات سهلة يمكن حلها بالقوانين
عن طريق فرض دالة مشتقتها موجودة بالسؤال.



انتبه



بعد الفرض يجب تول السؤال بأكملة الى
صيغة المتغير الجديد

Example

$$\int 2x(x^2 + 10)^{55} dx$$

Let $u = x^2 + 10$

$du = 2x dx$

فرض الدالة الي عندي مشتقتها

اشتقاق الفرض

اشيل وأبدل

$\int u^{55} du$

$\frac{u^{56}}{56} + C$

$\frac{(x^2 + 10)^{56}}{56} + C$

لازم بعد الفرض كل السؤال يتحول الى

u

If not

الاشتقاق خاطئ

Or

الفرض خاطئ



طريقة الفرض



وين الدالة الي عدي مشتقتها؟؟

للتسهيل !!

u

$\sin$ (function)

$\cos$ (function)

$\tan$ (function)

.....

e^{function}

a^{function}

$u = \text{function}$

بسط

مقام

$u = \text{المقام}$

!!
المليشيق
الأس

ما داخل الأس المليق u

$$\int 3x^2(x^3 - 2)^5 dx$$

$\int 3x^2 (x^3 - 2)^5 dx$
 Let $u = x^3 - 2$
 $du = 3x^2 dx$

$$\int u^5 du$$

$$\frac{u^6}{6} + C = \frac{(x^3 - 2)^6}{6} + C$$

$$\int x \sqrt{(x^2 + 7)^3} dx$$

$\int x (x^2 + 7)^{\frac{3}{2}} dx$
 Let $u = x^2 + 7$
 $du = 2x dx$
 $\frac{1}{2} du = x dx$

$$\frac{1}{2} \int u^{\frac{3}{2}} du$$

$$\frac{1}{2} \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + C = \frac{(x^2 + 7)^{\frac{5}{2}}}{5} + C$$

$$\int e^x \sqrt{1+e^x} dx$$

$$\int e^x (1+e^x)^{\frac{1}{2}} dx$$

Let $u = 1+e^x$
 $du = e^x dx$

$$\int u^{\frac{1}{2}} du$$

$$= \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3} (1+e^x)^{\frac{3}{2}} + C$$

$$\int \frac{e^x}{(1+e^x)^2} dx$$

$$\int e^x (1+e^x)^{-2} dx$$

Let $u = 1+e^x$
 $du = e^x dx$

$$\int u^{-2} du$$

$$= \frac{u^{-1}}{-1} + C = -\frac{1}{u} + C = -\frac{1}{1+e^x} + C$$

$$\int 4^x (4^x + 4)^4 dx$$

Let $u = 4^x + 4$
 $\div \ln 4 \quad du = 4^x \ln 4 \, dx$
 $\frac{1}{\ln 4} du = 4^x dx$

$$\frac{1}{\ln 4} \int u^4 du$$

$$\frac{1}{\ln 4} \frac{u^5}{5} + C \Rightarrow \frac{1}{\ln 4} \frac{(4^x + 4)^5}{5} + C$$

$$\int \frac{2^x}{\sqrt{2^x + 3}} dx$$

$$\int 2^x (2^x + 3)^{-\frac{1}{2}} dx$$

Let $u = 2^x + 3$
 $\div \ln 2 \quad du = 2^x \ln 2 \, dx$
 $\frac{1}{\ln 2} du = 2^x dx$

$$\frac{1}{\ln 2} \int u^{-\frac{1}{2}} du$$

$$\frac{1}{\ln 2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C = \frac{1}{\ln 2} \frac{(2^x + 3)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$\int \frac{(1 + \sqrt{x})^5}{\sqrt{x}} dx$$

Let $u = 1 + \sqrt{x}$

$$du = \frac{1}{2\sqrt{x}} dx \longrightarrow 2du = \frac{1}{\sqrt{x}} dx$$

$$2 \int u^5 du$$

$$2 \frac{u^6}{6} + C = \frac{1}{3} (1 + \sqrt{x})^6 + C$$

$$\int \frac{1}{\sqrt[3]{x^2} (7 + \sqrt[3]{x})^7} dx$$

$$\int x^{-\frac{2}{3}} (7 + x^{\frac{1}{3}})^{-7} dx$$

Let $u = 7 + x^{\frac{1}{3}}$

$$du = \frac{1}{3} x^{-\frac{2}{3}} dx \longrightarrow 3du = x^{-\frac{2}{3}} dx$$

$$3 \int u^{-7} du$$

$$3 \frac{u^{-6}}{-6} + C = -\frac{1}{2} (7 + x^{\frac{1}{3}})^{-6} + C$$

$$\int \frac{\left(7 + \frac{1}{x^2}\right)^9}{x^3} dx$$

$$\int (7 + x^{-2})^9 x^{-3} dx$$

Let $u = 7 + x^{-2}$

$$du = -2x^{-3} dx \longrightarrow -\frac{1}{2} du = x^{-3} dx$$

$$-\frac{1}{2} \int u^9 du$$

$$-\frac{1}{2} \frac{u^{10}}{10} + C = -\frac{1}{20} (7 + x^{-2})^{10} + C$$

$$\int \frac{\sec^2 x}{(\tan x + 3)^4} dx$$

$$\int \sec^2 x (\tan x + 3)^{-4} dx$$

Let $u = \tan x + 3$

$$du = \sec^2 x dx$$

$$\int u^{-4} du$$

$$= \frac{u^{-3}}{-3} + C = \frac{(\tan x + 3)^{-3}}{-3} + C$$

Evaluate the integral

Dec 2017

$$\int \frac{1}{\sqrt{x}(\sqrt{x}+1)^2} dx$$

Let $u = \sqrt{x} + 1$

$$du = \frac{1}{2\sqrt{x}} dx \rightarrow 2du = \frac{1}{\sqrt{x}} dx$$

$$2 \int \frac{1}{u^2} du$$

$$2 \int u^{-2} du$$

$$2 \frac{u^{-1}}{-1} + C$$

$$-2 \frac{1}{u} + C$$

$$-2 \frac{1}{\sqrt{x} + 1} + C$$

$$\int \frac{2x}{(x^2 + 5)^3} dx$$

$$\int 2x (x^2 + 5)^{-3} dx$$

$$\text{Let } u = x^2 + 5$$

$$du = 2x dx$$

$$\int u^{-3} du = \frac{u^{-2}}{-2} + C$$

$$= \frac{(x^2 + 5)^{-2}}{-2} + C$$

$$\int \frac{\sin x}{(\cos x - 3)^2} dx$$

$$\int \sin x (\cos x - 3)^{-2} dx$$

$$\text{Let } u = \cos x - 3$$

$$du = -\sin x dx$$

$$-du = \sin x dx$$

$$-\int u^{-2} du$$

$$= -\frac{u^{-1}}{-1} + C = \frac{1}{u} + C = \frac{1}{\cos x - 3} + C$$

$$\int \frac{1 - \sin x}{\sqrt{x + \cos x}} dx$$

$$\int (1 - \sin x) (x + \cos x)^{-\frac{1}{2}} dx$$

Let $\begin{cases} u = x + \cos x \\ du = (1 - \sin x) dx \end{cases}$

$$\int u^{-\frac{1}{2}} du$$

$$= \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C = 2 (x + \cos x)^{\frac{1}{2}} + C$$

$$\int \frac{\sinh x}{(\cosh x + 3)^7} dx$$

$$\int \sinh x (\cosh x + 3)^{-7} dx$$

Let $\begin{cases} u = \cosh x + 3 \\ du = \sinh x dx \end{cases}$

$$\int u^{-7} du$$

$$= \frac{u^{-6}}{-6} + C = \frac{(\cosh x + 3)^{-6}}{-6} + C$$

$$\int \frac{(1 + \cot 7x)^9}{\sin^2 7x} d\theta$$

$$\int \csc^2 7x (1 + \cot 7x)^9 dx$$

Let $u = 1 + \cot 7x$

$$du = -7 \csc^2 7x dx$$

$$-\frac{1}{7} du = \csc^2 7x dx$$

$$-\frac{1}{7} \int u^9 du$$

$$-\frac{1}{7} \frac{u^{10}}{10} + C = -\frac{1}{70} (1 + \cot 7x)^{10} + C$$

$$\int \left(\frac{\tan \theta + 7}{\cos \theta} \right)^2 d\theta$$

$$\int \frac{(\tan \theta + 7)^2}{\cos^2 \theta} d\theta$$

$$\int \sec^2 \theta (\tan \theta + 7)^2 d\theta$$

Let $u = \tan \theta + 7$
 $du = \sec^2 \theta d\theta$

$$\int u^2 du$$

$$\frac{u^3}{3} + C = \frac{(\tan \theta + 7)^3}{3} + C$$

Evaluate the integral

Aug 2017

$$\int \sin^3(x) \cos(x) dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$\int u^3 du$$

$$= \frac{u^4}{4} + C$$

$$= \frac{(\sin x)^4}{4} + C$$

Evaluate the integral

May 2018

$$\int 2 \cos(2x) \sqrt{1 + \sin(2x)} dx$$

$$\int 2 \cos(2x) (1 + \sin(2x))^{\frac{1}{2}} dx$$

Let $u = 1 + \sin(2x)$
 $du = 2 \cos(2x) dx$

$$\int u^{\frac{1}{2}} du$$

$$= \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{2}{3} (1 + \sin(2x))^{\frac{3}{2}} + C$$

Evaluate the integral

May 2017

$$\int \frac{\cos(\pi x)}{(1 + \sin(\pi x))^3} dx$$

$$\int \cos(\pi x) (1 + \sin(\pi x))^{-3} dx$$

$$\text{Let } u = 1 + \sin(\pi x)$$

$$du = \pi \cos(\pi x) dx \rightarrow \frac{1}{\pi} du = \cos(\pi x) dx$$

$$\frac{1}{\pi} \int u^{-3} du$$

$$\frac{1}{\pi} \frac{u^{-2}}{-2} + C$$

$$\frac{-1}{2\pi} \frac{1}{u^2} + C$$

$$\frac{-1}{2\pi} \frac{1}{(1 + \sin(\pi x))^2} + C$$

$$\int \frac{2\tan x + 1}{\cos^2 x \sqrt[3]{\tan^2 x + \tan x + 5}} dx$$

$$\int \sec^2 x (\tan^2 x + \tan x + 5)^{-\frac{1}{3}} (2\tan x + 1) dx$$

Let $u = \tan^2 x + \tan x + 5$

$$du = (2\tan x \sec^2 x + \sec^2 x) dx$$

$$du = \sec^2 x (2\tan x + 1) dx$$

$$\int u^{-\frac{1}{3}} du$$

$$\frac{u^{\frac{2}{3}}}{\frac{2}{3}} + C$$

$$\frac{3}{2} (\tan^2 x + \tan x + 5)^{\frac{2}{3}} + C$$

$$\int (1 + \cos^2 x)^3 \sin 2x \, dx$$

Let $u = 1 + \cos^2 x$

$$du = 2 \cos x (-\sin x) \, dx$$

$$du = -2 \cos x \sin x \, dx$$

$$du = -\sin 2x \, dx$$

$$\sin 2x = 2 \sin x \cos x$$

* -1

$$-du = \sin 2x \, dx$$

$$-\int u^3 \, du$$

$$-\frac{u^4}{4} + C = -\frac{(1 + \cos^2 x)^4}{4} + C$$



$$\int (1 + \cos^2 2x)^3 \sin 4x \, dx$$



Let $u = 1 + \cos^2 2x$

$$du = 2 \cos 2x (-2 \sin 2x) \, dx$$

$$\div (-4)$$

$$= -4 \cos 2x \sin 2x \, dx$$

$$\sin 2x = 2 \sin x \cos x$$

$$\sin 4x = 2 \sin 2x \cos 2x$$

$$\frac{1}{2} \sin 4x = \sin 2x \cos 2x$$

$$du = -4 \frac{1}{2} \sin 4x \, dx$$

$$\div (-2) \, du = -2 \sin 4x \, dx$$

$$\frac{1}{2} du = \sin 4x \, dx$$

$$\Rightarrow -\frac{1}{2} \int u^3 \, du = -\frac{1}{2} \frac{u^4}{4} + C$$

المراد
المراد

$$\int \frac{\sqrt{u^6 + 5u^4}}{u} du$$

$$\int \frac{\sqrt{u^4(u^2+5)}}{u} du$$

$$\int \frac{\sqrt{u^4} \sqrt{u^2+5}}{u} du$$

$$\int \frac{u^2 \sqrt{u^2+5}}{u} du$$

$$\int u (u^2+5)^{\frac{1}{2}} du$$

Let $t = u^2 + 5$

$dt = 2u du \rightarrow \frac{1}{2} dt = u du$

$$\frac{1}{2} \int t^{\frac{1}{2}} dt$$

$$\frac{1}{2} \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{(u^2 + 5)^{\frac{3}{2}}}{3} + C$$

قبل الحل
يمكن
التبسيط
بالعامل
المشترك

$$\int \sqrt{8x+1} dx$$

$$\int (8x+1)^{\frac{1}{2}} dx$$

Let $u = 8x+1$

$$du = 8 dx$$

$$\frac{1}{8} du = dx$$

$$\frac{1}{8} \int u^{\frac{1}{2}} du = \frac{1}{8} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{1}{8} \frac{(8x+1)^{\frac{3}{2}}}{\frac{3}{2}} + C$$



$$\int x\sqrt{x+1} dx$$



$$\int x(x+1)^{\frac{1}{2}} dx$$

Let $u = x+1$

$$du = dx$$

$$\rightarrow x = u - 1$$

$$\int (u-1)u^{\frac{1}{2}} du$$

$$\int (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du$$

$$\int (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du$$

$$= \frac{u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{(x+1)^{\frac{5}{2}}}{\frac{5}{2}} - \frac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}} + C$$

Evaluate the integral

Jan 2017

$$\int_0^{\frac{\pi}{4}} 4(1 + \tan x)^2 \sec^2 x \, dx$$

Let $u = 1 + \tan x$

$$du = \sec^2 x \, dx$$

$$\int 4 u^2 \, du$$

$$= 4 \frac{u^3}{3} + C$$

$$= \frac{4}{3} \left[(1 + \tan x)^3 \right]_0^{\frac{\pi}{4}} + C$$

$$= \frac{4}{3} \left[(1+1)^3 - (1+0)^3 \right]$$

$$= \frac{4}{3} [8 - 1]$$

$$= \frac{4}{3} (7)$$

$$= \frac{28}{3}$$

$$\int_{-1}^1 \frac{x}{\sqrt{5-x^2}} dx$$



الأعلى



$$f(x) = \frac{x}{\sqrt{5-x^2}}$$

$$\begin{aligned} f(-x) &= \frac{-x}{\sqrt{5-(-x)^2}} \\ &= \frac{-x}{\sqrt{5-x^2}} \\ &= -f(x) \end{aligned}$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-1}^1 \frac{x}{\sqrt{5-x^2}} dx = 0$$

$$\int_{-1}^1 x (5-x^2)^{-\frac{1}{2}} dx$$

$$\begin{aligned} \text{Let } u &= 5-x^2 \\ du &= -2x dx \\ -\frac{1}{2} du &= x dx \end{aligned}$$

$$= -\frac{1}{2} \int u^{-\frac{1}{2}} du$$

$$= -\frac{1}{2} \cdot \frac{u^{\frac{1}{2}}}{\frac{1}{2}}$$

$$= - \left[(5-x^2)^{\frac{1}{2}} \right]_{-1}^1$$

$$= - \left[4^{\frac{1}{2}} - 4^{\frac{1}{2}} \right]$$

0

$$\int_{-2}^2 \left[\cos(\pi x) - \frac{x^3}{\sqrt{1+x^4}} \right] dx$$

$$\int_{-2}^2 \cos(\pi x) dx - \int_{-2}^2 \frac{x^3}{\sqrt{1+x^4}} dx$$

$$\left[\frac{\sin(\pi x)}{\pi} \right]_{-2}^2$$

$$\frac{1}{\pi} [\sin(2\pi) - \sin(-2\pi)]$$

$$\frac{1}{\pi} [\sin(2\pi) + \sin(2\pi)]$$

$$\frac{1}{\pi} [0 + 0] = 0$$

Even حل هدفه طلعت صفر

$$f(x) = \cos(\pi x)$$

$$f(-x) = \cos(-\pi x) = \cos \pi x = f(x)$$

$\therefore f(x)$ is even function

$$\therefore 2 \int_0^2 \cos(\pi x) dx$$

$$2 \left[\frac{\sin \pi x}{\pi} \right]_0^2$$

$$= \frac{2}{\pi} [\sin 2\pi - 0] = 0$$

$$\text{Let } f(x) = \frac{x^3}{\sqrt{1+x^4}}$$

$$f(-x) = \frac{(-x)^3}{\sqrt{1+(-x)^4}}$$

$$= \frac{-x^3}{\sqrt{1+x^4}}$$

$$= -f(x)$$

$\therefore f(x)$ is odd function

$$\therefore \int_{-2}^2 \frac{x^3}{\sqrt{1+x^4}} dx = 0$$

$$\therefore \int_{-2}^2 \left[\cos(\pi x) - \frac{x^3}{\sqrt{1+x^4}} \right] dx = 0 - 0 = 0$$

$$\int_0^3 \frac{x}{\sqrt{16+x^2}} dx$$

معيبه لو
طبيعتها
ODD

$$\int_0^3 x (16+x^2)^{-\frac{1}{2}} dx$$

Let $u = 16 + x^2$

$$du = 2x dx \longrightarrow \frac{1}{2} du = x dx$$

$$\frac{1}{2} \int u^{-\frac{1}{2}} du$$

$$\frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}}$$

$$\left[(16+x^2)^{\frac{1}{2}} \right]_0^3$$

$$(16+9)^{\frac{1}{2}} - (16+0)^{\frac{1}{2}}$$

$$(25)^{\frac{1}{2}} - (16)^{\frac{1}{2}}$$

$$5 - 4 = 1$$

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$$

Let $u = \frac{1}{x}$

$$du = -\frac{1}{x^2} dx \longrightarrow -du = \frac{1}{x^2} dx$$

$$-\int e^u du$$

$$-e^u = -\left[e^{\frac{1}{x}}\right]_1^2$$

$$= -\left[e^{\frac{1}{2}} - e^1\right]$$

$$\int_0^1 x e^{-x^2} dx$$

Let $u = -x^2$

$$du = -2x dx \longrightarrow -\frac{1}{2} du = x dx$$

$$-\frac{1}{2} \int e^u du$$

$$-\frac{1}{2} e^u = -\frac{1}{2} \left[e^{-x^2}\right]_0^1$$

$$= -\frac{1}{2} \left[e^{-1} - \underset{e^0}{1}\right]$$

$$\int \frac{\cos(\sin^{-1}x)}{\sqrt{1-x^2}} dx$$

Let $u = \sin^{-1}x$

$$du = \frac{1}{\sqrt{1-x^2}} dx$$

$$\int \cos u \, du$$

$$= \sin u + C$$

$$= \sin \sin^{-1} x + C$$

$$= x + C$$

$$\int x^4 \tan^2 x^5 \, dx$$

Let $u = x^5$

$$du = 5x^4 dx \longrightarrow \frac{1}{5} du = x^4 dx$$

$$\frac{1}{5} \int \tan^2 u \, du$$

$$\frac{1}{5} \int (\sec^2 u - 1) \, du$$

$$\frac{1}{5} [\tan u - u] + C$$

$$\frac{1}{5} [\tan x^5 - x^5] + C$$

$$\int \frac{ds}{\sqrt{s} \cos^2 \sqrt{s}}$$

Let $u = \sqrt{s}$

$$du = \frac{1}{2\sqrt{s}} ds \longrightarrow 2du = \frac{1}{\sqrt{s}} ds$$

$$= 2 \int \frac{1}{\cos^2 u} du$$

$$= 2 \int \sec^2 u du$$

$$= 2 \tan u + C$$

$$= 2 \tan \sqrt{s} + C$$

$$\int \frac{\sec(\ln x) \sin(\ln x)}{x \cos(\ln x)} dx$$

Let $u = \ln x$

$$du = \frac{1}{x} dx$$

$$= \int \frac{\sec u \sin u}{\cos u} du$$

$$= \int \sec u \tan u du$$

$$= \sec u + C$$

$$= \sec(\ln x) + C$$



Tricky question !!



Evaluate the integral

Jan 2017

$$\int (\cos x)(\sin(\sin x)) dx$$

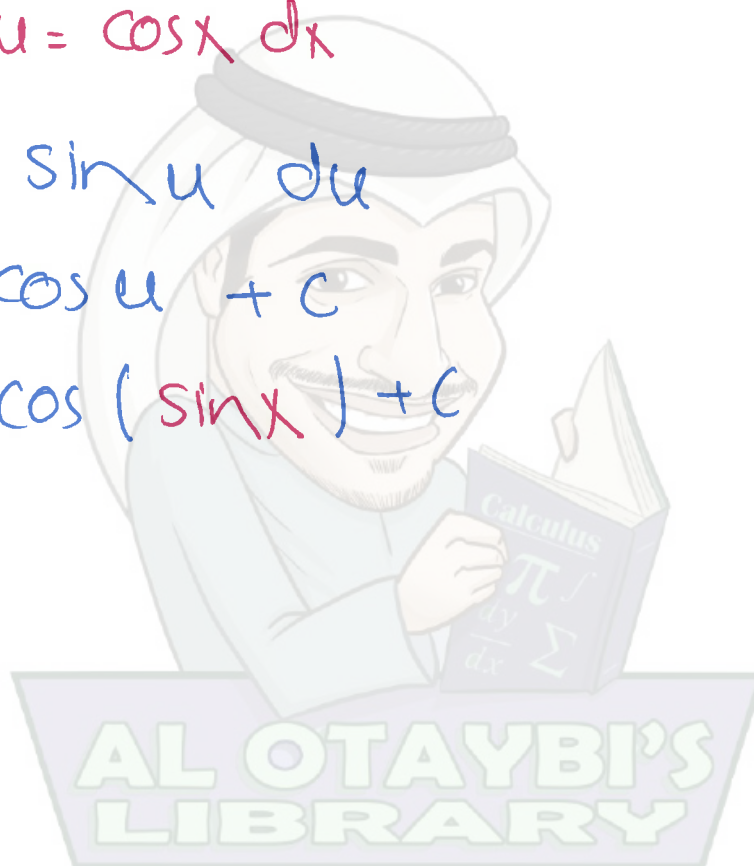
Let $u = \sin x$

$$du = \cos x dx$$

$$\int \sin u du$$

$$= -\cos u + C$$

$$= -\cos(\sin x) + C$$





$$\int_0^{\frac{\pi}{2}} \cos(\sin x) \cos x \, dx$$



Let $u = \sin x$
 $du = \cos x \, dx$

$$\int \cos u \, du$$

$\sin u$ ~~$\times$~~

$$\left[\sin(\sin x) \right]_0^{\frac{\pi}{2}}$$

$$\sin\left(\sin \frac{\pi}{2}\right) - \sin(\sin 0)$$

$$\sin(1) - \sin(0)$$

$$\sin(1) - 0$$

$$\sin(1)$$

$$\int \cos(7x - 5) dx$$



الأحلى



* فالة راعيا *

قانون

$$\text{Let } u = 7x - 5$$

$$du = 7 dx$$

$$\frac{1}{7} du = dx$$

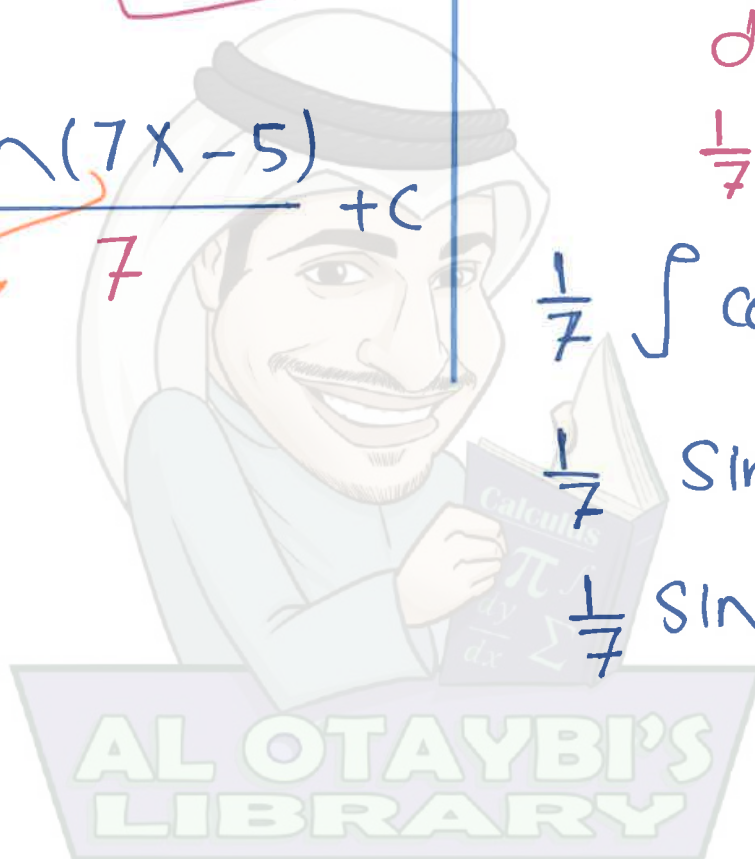
$$= \frac{\sin(7x - 5)}{7} + C$$

معامل x

$$\frac{1}{7} \int \cos u du$$

$$\frac{1}{7} \sin u + C$$

$$\frac{1}{7} \sin(7x - 5) + C$$



$$\int \frac{5 - 4x^3}{x^6} dx + \int x^{-\frac{3}{4}} \cos\left(2 + x^{\frac{1}{4}}\right) dx$$

$$\int \left(\frac{5}{x^6} - \frac{4x^3}{x^6} \right) dx + \int x^{-\frac{3}{4}} \cos(2 + x^{\frac{1}{4}}) dx$$

$$\int (5x^{-6} - 4x^{-3}) dx$$

$$= 5 \frac{x^{-5}}{-5} - 4 \frac{x^{-2}}{-2} + C_1$$

$$\text{Let } u = 2 + x^{\frac{1}{4}}$$

$$du = \frac{1}{4} x^{-\frac{3}{4}} dx$$

$$4 du = x^{-\frac{3}{4}} dx$$

$$4 \int \cos u du$$

$$4 \sin u + C_2$$

$$4 \sin(2 + x^{\frac{1}{4}}) + C_2$$

$$\Rightarrow 5 \frac{x^{-5}}{-5} - 4 \frac{x^{-2}}{-2} + 4 \sin(2 + x^{\frac{1}{4}}) + C$$

$$\int \frac{\cos x}{\sin x + 5} dx$$

Let $u = \sin x + 5$
 $du = \cos x dx$

$$\int \frac{1}{u} du$$

$$= \ln |u| + C$$

$$= \ln |\sin x + 5| + C$$

$$\int \frac{e^x}{e^x - 2} dx$$

Let $u = e^x - 2$
 $du = e^x dx$

$$\int \frac{1}{u} du$$

$$= \ln |u| + C$$

$$= \ln |e^x - 2| + C$$

$$\int \frac{x+2}{x^2+4x} dx$$

Let $u = x^2 + 4x$

$$du = (2x + 4) dx$$

$$\div 2$$

$$\frac{1}{2} du = (x + 2) dx$$

$$\int \frac{1}{u} \cdot \frac{1}{2} du$$

$$\frac{1}{2} \int \frac{1}{u} du$$

$$\frac{1}{2} \ln |u| + C$$

$$\frac{1}{2} \ln |x^2 + 4x| + C$$

$$\int \frac{5x}{x^2 + 7} dx$$

Let $u = x^2 + 7$

$$du = 2x dx$$

$$\rightarrow \frac{1}{2} du = x dx$$

$$\frac{1}{2} \int \frac{5}{u} du$$

$$\frac{5}{2} \int \frac{1}{u} du$$

$$\frac{5}{2} \ln |u| + C$$

$$\frac{5}{2} \ln |x^2 + 7| + C$$

Evaluate the integral

July 2018

$$\int_0^{\frac{\pi}{4}} \frac{2\sec^2 x}{1 + \tan x} dx$$

Let $u = 1 + \tan x$
 $du = \sec^2 x dx$

$$\begin{aligned} & \int \frac{2}{u} du \\ &= 2 \ln|u| = 2 \left[\ln|1 + \tan x| \right]_0^{\frac{\pi}{4}} \\ &= 2 \left[\ln|1+1| - \ln|1+0| \right] = 2 \ln 2 \end{aligned}$$

Evaluate the integral

May 2018

$$\int_1^2 \frac{x+2}{x^2+4x+4} dx$$

Let $u = x^2 + 4x + 4$
 $du = (2x+4) dx$
 $du = 2(x+2) dx \longrightarrow \frac{1}{2} du = (x+2) dx$

$$\begin{aligned} & \frac{1}{2} \int \frac{1}{u} du \\ & \frac{1}{2} \ln|u| = \frac{1}{2} \left[\ln|x^2+4x+4| \right]_1^2 \\ &= \frac{1}{2} \left[\ln|4+8+4| - \ln|1+4+4| \right] = \frac{1}{2} [\ln 16 - \ln 9] \end{aligned}$$

$$\int \frac{1+x}{1+x^2} dx$$

$$\int \left[\frac{1}{1+x^2} + \frac{x}{1+x^2} \right] dx$$

$$\int \frac{1}{1+x^2} dx + \int \frac{x}{1+x^2} dx$$

$$\tan^{-1} x + C_1$$

Let $u = 1+x^2$
 $du = 2x dx$
 $\frac{1}{2} du = x dx$

$$\frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln|u| + C_2$$

$$= \frac{1}{2} \ln|1+x^2| + C_2$$

$$\tan^{-1} x + \frac{1}{2} \ln|1+x^2| + C$$

$$\int \frac{\operatorname{sech} x (1 - x \tanh x)}{x \operatorname{sech} x + 1} dx$$

Let $u = x \operatorname{sech} x + 1$

الضرب في $\frac{1}{\cosh x}$ + المشتق
 $du = (1) \operatorname{sech} x + x [-\operatorname{sech} x \tanh x] dx$
 $du = \operatorname{sech} x (1 - x \tanh x) dx$

$$= \int \frac{1}{u} du$$

$$= \ln|u| + C$$

$$= \ln|x \operatorname{sech} x + 1| + C$$



$$\int \frac{e^x}{e^{2x} + 1} dx$$



$$\int \frac{e^x}{(e^x)^2 + 1} dx$$

Let $u = e^x$
 $du = e^x dx$

$$= \int \frac{1}{u^2 + 1} du$$

$$= \tan^{-1} u + C = \tan^{-1} e^x + C$$



$$\int \frac{x}{1 + x^4} dx$$



$$\int \frac{x}{1 + (x^2)^2} dx$$

Let $u = x^2$
 $du = 2x dx \rightarrow \frac{1}{2} du = x dx$

$$\frac{1}{2} \int \frac{1}{1 + u^2} du$$

$$\frac{1}{2} \tan^{-1} u + C$$

$$\frac{1}{2} \tan^{-1} (x^2) + C$$

$$\int \frac{\tan^{-1}x}{x^2 + 1} dx$$

Let $u = \tan^{-1}x$

$$du = \frac{1}{x^2 + 1} dx$$

$$\int u du$$

$$\frac{u^2}{2} + C = \frac{(\tan^{-1}x)^2}{2} + C$$

$$\int \frac{(\ln x)^2}{x} dx$$

Let $u = \ln x$

$$du = \frac{1}{x} dx$$

$$\int u^2 du$$

$$\frac{u^3}{3} + C = \frac{(\ln x)^3}{3} + C$$

$$\int \frac{\sqrt{\sec^{-1}x}}{x\sqrt{x^2-1}} dx$$

Let $u = \sec^{-1}x$

$$du = \frac{1}{x\sqrt{x^2-1}} dx$$

$$\int \sqrt{u} du = \int u^{\frac{1}{2}} du$$

$$= \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3} (\sec^{-1}x)^{\frac{3}{2}} + C$$

$$\int \frac{1}{(\sin^{-1}x)^2 \sqrt{1-x^2}} dx$$

Let $u = \sin^{-1}x$

$$du = \frac{1}{\sqrt{1-x^2}} dx$$

$$\int \frac{1}{u^2} du = \int u^{-2} du$$

$$= \frac{u^{-1}}{-1} + C = \frac{-1}{u} + C$$

$$= \frac{-1}{\sin^{-1}x} + C$$

Evaluate the integral

$$\int \frac{dx}{x(\ln x)^5}$$

July 2018

Let $u = \ln x$

$$du = \frac{1}{x} dx$$

$$\int \frac{1}{u^5} du$$

$$\int u^{-5} du$$

$$= \frac{u^{-4}}{-4} + C$$

$$= \frac{-1}{4 u^4} + C$$

$$= \frac{-1}{4 (\ln x)^4} + C$$

Evaluate the integral

$$\int_1^e \frac{\ln(x^2)}{x} dx$$

May 2017

$$\int_1^e \frac{2 \ln x}{x} dx$$

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\int 2u du = 2 \frac{u^2}{2}$$

$$= \left[(\ln x)^2 \right]_1^e$$

$$= (\ln e)^2 - (\ln 1)^2 = (1)^2 - 0 = 1$$

Evaluate the integral

$$\int_e^{e^2} \frac{dx}{x \ln x}$$

Aug 2017

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\int \frac{1}{u} du$$

$$= \ln |u|$$

$$= \left[\ln |\ln x| \right]_e^{e^2}$$

$$= \ln |\ln e^2| - \ln |\ln e|$$

$$= \ln |2| - \ln |1|$$

$$= \ln 2 - 0$$

$$= \ln 2$$

The slope of the curve $y = f(x)$ is given by

$$f'(x) \quad \swarrow \quad m = x \cos(x^2) + 1$$

Find $f(x)$ knowing that this curve passes through the point $(0, 1)$.

$$f'(x) = x \cos(x^2) + 1$$

$$f(x) = \int (x \cos(x^2) + 1) dx$$

$$\int x \cos(x^2) dx + \int 1 dx$$

Let $u = x^2$
 $du = 2x dx$
 $\frac{1}{2} du = x dx$

$$X + C_2$$

$$\frac{1}{2} \int \cos u du$$

$$\frac{1}{2} \sin u + C_1$$

$$\frac{1}{2} \sin(x^2) + C_1$$

$$\Rightarrow f(x) = \frac{1}{2} \sin(x^2) + x + C$$

at $(0, 1)$

$$1 = \frac{1}{2} \sin 0 + 0 + C \Rightarrow C = 1$$

$$\Rightarrow f(x) = \frac{1}{2} \sin(x^2) + x + 1$$